

PROJECT FORMULATION

Concept of Project Formulation

Project formulation is the systematic development of a project idea for the eventual objective of arriving at an investment decision. It involves a step-by-step investigation and development of project idea. And it provides a controlled mechanism for restricting expenditure on project development.

According to **B.B. Goel**, Project formulation refers to a series of steps to be taken to convert an idea or aspiration into a feasible plan of action.

According **G. Myrdal**, Project formulation is one of the basic techniques through which planning can be changed from an idea to an institutional and rational base.

Project formulation is a process involving the joint efforts of a team of experts. Each member of the team should be familiar with the broad strategy, objectives and other ingredients of the project. The government official who deals with the project's final clearance has to be treated as forming part of the team. A well-formulated feasibility report provides a medium, which cuts across scientific, social and positional prejudices and provides a common meeting ground for all those who have a contribution to make in successful implementation of a project.

Project team should consist of experts in major substantive fields of the project. Depending on the situation any large project should comprise the following team members.

- a. One industrial economist
- b. One market analyst
- c. One or more technologist/engineer specializing in the appropriate industry.
- d. One mechanical and/or industrial engineer.
- e. One management accounting expert.

Stages of Project Formulation

1. Project identification

Project identification is concerned with the collection, compilation and analysis of economic data for the eventual purpose of locating possible opportunities for investment and with the development of the characteristics of such opportunities. Opportunities, according to **Drucker (1955)**, are of three kinds; complementary, break-through and additive.

- i) Complementary opportunities involve the introduction of new ideas and as such do lead to a certain amount of change in the existing structure.
- ii) Break-through opportunities, on the other hand, involve fundamental changes in both the structure and character of business.
- iii) Additive opportunities are those opportunities, which enable the decision-maker to better utilize the existing resources without involving any change in the character of business. Additive

opportunities involve the least amount of disturbance to the existing state of affairs and hence the least amount of risk.

The element of risk is more in other two opportunities. When the element of risk increases, it becomes more important to precisely define the scope and nature of project idea, to develop alternative solutions for achieving the project objectives and to select the best possible approach so as to minimize both resource consumption and risks and to optimize the return or gains.

Project identification cannot be complete without identifying the characteristics of a project. Every project has three basic dimensions-inputs, outputs and social costs and benefits. The input characteristics define what the project will consume in terms of raw materials, energy, manpower, finance and organizational setup. The nature and magnitude of each of these inputs must be determined in order to make the input characteristic explicit.

The output characteristics of a project define what the project will generate in the form of goods and services, employment, revenue etc., the quantity and quality of all these outputs should be clearly specified.

In addition to inputs and outputs every project has an impact on the society. It inevitably affects the current equilibriums of the demand and supply in the economy. It is necessary to evaluate carefully the sacrifice, which the society will be required to make, and the benefits that will accrue to the society from a given project.

Project do not emerge themselves. The inputs to set up a project can come from different sources such as Government agencies, credit and financial institutions, non-governmental organization like chambers of commerce and industry, institutional groups, technical consultancy organizations and international collaborations. Once the venture ideas have been developed by entrepreneurs by following on or combination of sources explained, these have to be screened and evaluated in a preliminary fashion on the basis of internal and external constraints prior to being put to additional tests of pre-feasibility. This project identification comes to an end by laying down specific project objectives clearly and concisely and without any ambiguity so that these convey one and the same meaning to all concerned.

2. Technical Analysis

Technical analysis includes the following:

1. Input analysis

a. Site: Location of enterprise; owned or leasehold land; industrial area; No Objection Certificate from the Municipal Authorities if the enterprise location falls in the residential area.

b. Raw Material: Requirement of raw material, whether inland or imported, sources of raw material supply.

c. Human Resource: Availability of skilled, Semi-skilled labor in the area, arrangements for training laborers in various skills.

d. Technology Selected: The selection of technology, arrangements made for acquiring it should be mentioned in the business plan.

e. Utilities: These include: **Power:** Requirement for power, load sanctioned, availability of power. **Fuel:** Requirement for fuel items such as coal, coke, oil or gas, state of their availability. **Water:** The sources and quality of water should be clearly stated in the project report.

f. Pollution Control: The aspects like scope of dumps; sewage system and sewage treatment plant should be clearly stated in case of industries producing emissions.

g. Communication System: Availability of communication facilities, e.g., telephone, fax, internet facility, etc. should be stated in the project report.

h. Transport Facilities: Requirements for transport, mode of transport, potential means of transport, distances to be covered, bottlenecks etc., should be stated in the business plan.

i. Other Common Facilities: Availability of common facilities like machine shops, welding shops and electrical repair shops etc. should be stated in the report.

j. Production Process: A mention should be made for process involved in production and period of conversion from raw material into finished goods.

k. Machinery and Equipment: A complete list of items of machinery and equipments required indicating their size, type, cost and sources of their supply should be enclosed with the project report.

l. Capacity of the Plant: The installed licensed capacity of the plant along with the shifts should also be mentioned in the project report.

m. Research and Development: A mention should be made in the project report regarding proposed research and development activities to be undertaken in future.

2. Demand and Supply analysis

State the total expected demand for the product and present supply position. This should also be mentioned how much of the gap will be filled up by the proposed unit. We can determine the demand of consumer's in following way

- a. Survey of buyer intentions;
- b. Composite of sales force opinion;
- c. Expert's opinion;
- d. Market test method;
- e. Statistical method.

3. Feasibility study

Feasibility studies aim to objectively and rationally uncover the strengths and weaknesses of an existing business or proposed venture, opportunities and threats as presented by the environment, the resources required to carry through, and ultimately the prospects for success. In its simplest terms, the two criteria to judge feasibility are cost required and value to be attained. As such, a well-designed feasibility study should provide a historical background of the business or project, description of the product or service, accounting statements, details of the operations and management, marketing research and policies, financial data, legal requirements and tax obligations. Generally, feasibility studies precede technical development and project implementation.

1. Management Feasibility study

Management appraisal is related to the technical and managerial competence, integrity, knowledge of the project, managerial competence of the promoters etc. The promoters should have the knowledge and ability to plan, implement and operate the entire project effectively. The past record of the promoters is to be appraised to clarify their ability in handling the projects.

2. Technical Feasibility study

Technical feasibility analysis is the systematic gathering and analysis of the data pertaining to the technical inputs required and formation of conclusion there from. The availability of the raw materials, power, sanitary and sewerage services, transportation facility, skilled man power, engineering facilities, maintenance, local people etc are coming under technical analysis. This feasibility analysis is very important since its significance lies in planning the exercises, documentation process, risk minimization process and to get approval.

3. Financial Feasibility study

One of the very important factors that a project team should meticulously prepare is the financial viability of the entire project. This involves the preparation of cost estimates, means of financing, financial institutions, financial projections, break-even point, ratio analysis etc. The cost of project includes the land and sight development, building, plant and machinery, technical know-how fees, pre-operative expenses, contingency expenses etc. The means of finance includes the share capital, term loan, special capital assistance, investment subsidy, margin money loan etc. The financial projections include the profitability estimates, cash flow and projected balance sheet. The ratio analysis will be made on debt equity ratio and current ratio. We can consider the following tools and techniques in financial feasibility

- a. Pay back period (PBP)
- b. Net Present Value (NPV)
- c. Benefit Cost ratio (BCR)
- d. Internal Rate of Return (IRR)
- e. Financial Ratios

4. Commercial Feasibility study

In the commercial appraisal many factors are coming. The scope of the project in market or the beneficiaries, customer friendly process and preferences, future demand of the supply, effectiveness of the selling arrangement, latest information availability in all areas, government control measures, etc. The appraisal involves the assessment of the current market scenario, which enables the project to get adequate demand. Estimation, distribution and advertisement scenario also to be here considered into.

5. Economic Feasibility study

How far the project contributes to the development of the sector, industrial development, social development, maximizing the growth of employment, etc. are kept in view while evaluating the economic feasibility of the project.

Cost-based study: It is important to identify cost and benefit factors, which can be categorized as follows: 1. Development costs; and 2. Operating costs. This is an analysis of the costs to be incurred in the system and the benefits derivable out of the system.

Time-based study: This is an analysis of the time required to achieve a return on investments. The future value of a project is also a factor.

6. Environmental Feasibility study

Environmental appraisal concerns with the impact of environment on the project. The factors include the water, air, land, sound, geographical location etc.

4. Project Appraisal

Project appraisal is the systematic and comprehensive review of the economic, environmental, financial, social, technical and other such aspects of a project to determine if it will meet its objectives.

If a project is well formulated and thoroughly appraised, a good follow-through on the subsequent stages of the project formulation will see to its goals being achieved. Appraisal involves a careful checking of the basic data, assumptions and methodology used in project preparation, an in-depth review of the work plan, cost estimates and proposed financing, an assessment of the project's organizational and management aspects, and finally the validity of the financial, economic and social benefits expected from the project. On the basis of such an assessment, a judgement is reached as to whether the project is technically sound, financially justified and viable from the point of view of the economy as a whole.

According to **B B Goel**, Project appraisal is a tool to examine as to whether in the given situation, it would be most realistic, reliable and reasonable one to commit resources or not.

According to **P.K. Motto**, Project appraisal is the process of evaluation the salient features of feasibility analysis, techno-economic analysis, design and network analysis, input analysis, financial analysis and social cost-benefit analysis of a project.

Financial institutions as well as promoters of projects are keenly interested in project appraisal. If the appraisal is unrealistic, then it is possible to drop a very good investment proposal and thereby lose a good investment opportunity. On the other hand if inadequate appraisal leads to investments in an unviable project, investments would not generate the expected returns and may lead to sickness of the project. It is, therefore, essential to consider all aspects of project viability; before deciding on further investments in the project.

The following specific aspects must be covered

- Technical Viability
- Commercial Viability - Market Demand
- Financial Viability
- Economic Viability
- Environmental Viability
- Managerial Viability

In essence, the project should be technically feasible and adequate demand should exist for selling the products generated by the operation of the project facilities. There should be adequate financial returns, and adequate returns to the national economy. All environmental requirements must be possible to be fulfilled and the enterprise must have the capability to manage the facilities efficiently and effectively after the project is over.

Problem of Project Formulation

The entrepreneur in a developing country has to encounter a number of problems while establishing a new project. These problems cause greater concern to many enthusiastic entrepreneurs. However, they could be saved to a greater extent by undertaking a project formulation exercise at the appropriate time.

1. Selection of appropriate technology

The first problem faced by an entrepreneur is in the matter of selection of appropriate technology for his enterprise. Modern technology developed in the highly industrialized countries may not be suitable for adoption in the developing countries as the conditions prevalent differ from country to country. For example, the optimal size of plants recommended for a highly industrialized country may be too big for acceptance in a developing country owing to the factors such as limited market for the products and limit availability of capital and skilled labor. Hence, the entrepreneur has to examine the project idea thoroughly as regards its design, production, and marketing after sales service, etc.

2. Influence of External Economics

The second problem relates to the absence or non-availability of external economics. No project can function in isolation in any economy. It has to depend on other industries for the supply of raw materials, power tools spare parts etc., or on ancillary enterprise which can provide technical, financial, and managerial services or a complex network of communication and transport facilities or an intricate system of business practices. The entrepreneur in developing countries is, therefore

to consider not only the basic costs of the project but also the ancillary costs, which in industrially advance countries would have been contributed by the external economics.

3. Dearth of Technically Qualified Personnel

The third problem is the non-availability of technically qualified and appropriate personnel. Modern technology calls for a certain minimum supply of various skills that are generally lacking in developing countries.

4. Knowledge about government regulations

Besides these problems the entrepreneur has to comprehend a number of Government directives. Import and export policies, price controls, etc. The difficulty is to be familiar with all these regulations, for they are not available in a consolidated and detailed form in most of the developing countries.

5. Product Selection

It is noticed that some entrepreneurs commit mistakes by selecting a wrong product for their enterprises. They select the product without giving due attention to product related other aspects such as size of the product markets, its future demand, competitive position, lifecycle, availability of required labor, raw material and technology. Hence, when you are selecting a product, take a comprehensive view.

6. Capacity Utilization Estimates

The entrepreneurs usually make over-optimistic estimates of capacity utilization. Their estimates are based on a completely false premise. The estimates are made in complete disregard of present-enterprise performance, prevailing market conditions, competitive atmosphere, the technical snags, etc. A business plan formulated as such falls prey to financial jugglery. Hence, avoid such temptations while estimating capacity utilization for your enterprise.

7. Market Study

Product production is ultimately meant for eventual sale. Hence, market study of the product assumes importance. Market study continues to be a grey area. But, there are some entrepreneurs who pass by this component of their business plan completely. Based on their nebulous ideas and scanty and scattered information on demand and supply of their proposed product, they conclude that market is just there waiting to be tapped. This is a wrong attitudinal block. Avoid it.

8. Location Selection

Most of the entrepreneurs select a location for their enterprises merely because it is their hometown or ancestral land or swayed by the Government offer of financial incentives and concessions to establish industries in a particular location; disregarding other factors like market proximity, availability of raw materials, manpower and infrastructural facilities.

9. Selection of Ownership Form

Many enterprises fail merely because the ownership form of enterprises is not suitable. Hence, select a suitable form of ownership taking a comprehensive view of the factors affecting the selection of a form of ownership.

Concept of Payback Period

One of the most commonly used techniques for evaluating investment proposal is the cash payback or payback period.

It attempts to calculate the period known as payback period required to recover the initial investment out of inflow of net cash flows or profit from the investment. In other words, it represents the number of years in which the investment is expected to 'pay for itself'

The length of time required to recover the cost of an investment. The payback period of a given investment or project is an important determinant of whether to undertake the position or project, as longer payback periods are typically not desirable for investment positions.

PBP can be Calculated as:

$$\text{PBP} = \frac{\text{Cost of investment}}{\text{Annual cash inflow}}$$

$$\text{or, PBP} = \frac{\text{NCO}}{\text{NCB}}$$

Where,

PBP = Payback period

NCO = Net cash outlay

NCB = Net cash benefit

All other things being equal, the better investment is the one with the shorter payback period. For example, if a project costs \$100,000 and is expected to return Tk. 20,000 annually, the payback period will be Tk. 100,000 / Tk. 20,000, or five years.

There are two main problems with the payback period method:

1. It ignores any benefits that occur after the payback period and, therefore, does not measure profitability.
2. It ignores the time value of money.

Because of these reasons, other methods of capital budgeting, like net present value, internal rate of return or discounted cash flow, are generally preferred.

ABC Enterprise has two mutually exclusive projects before it- Project X and Project Y. Both the projects require initial investment of Tk. 2,00,000.00. Net cash flow from each of the projects are as follows:

Year	Project X	Project Y
1	40000	30000
2	40000	70000
3	40000	40000
4	40000	47000
5	40000	32000
6	40000	30000

According to the pay back period technique, which project is acceptable?

Calculation of Project X

We know that,

$$\text{PBP} = \frac{\text{Cost of investment}}{\text{Annual cash inflow}}$$

$$\text{or, PBP} = \frac{\text{NCO}}{\text{NCB}}$$

$$\begin{aligned} \text{PBP} &= \frac{200000}{40000} \\ &= 5 \text{ Years} \end{aligned}$$

Where,

PBP = Payback period

NCO = Net cash outlay

NCB = Net cash benefit

Calculation of Project Y

Year	Project Y	Cumulative cash flow
1	30000	30000
2	70000	100000
3	40000	140000
4 = A	47000	187000 = C
5	32000 = D	219000
6	30000	249000

Again, We know that,

$$\begin{aligned} \text{PBP} &= A + \frac{\text{NCO} - C}{D} \\ &= 4 + \frac{200000 - 187000}{32000} \\ &= 4 + \frac{13000}{32000} \\ &= 4 + 0.40 \\ &= 4.40 \text{ Years} \end{aligned}$$

Where,

PBP = Payback period

NCO = Net cash outlay

A = Year in which cumulative cash inflow nearer to NCO

C = Cumulative cash inflow of the Year A

D = Cash inflow of the year following the year A

From the above calculation we can see that PBP of Project Y is less than Project X; so Project Y is acceptable.